# Quizrr Chapter-Wise Test for JEE Main - 2025

**Q.1**

Let $[x]$ denote the greatest integer $\leq x$. If $f(x)=[x]$ and $g(x)=|x|$, then the value of $f\left(g\left(\frac{8}{5}\right)\right)-g\left(f\left(-\frac{8}{5}\right)\right)$ is

**(A)** 2

**(B)** $-2$

**(C)** 1

**(D)** $-1$

**Q.2**

Let $\mathrm{f}:\{2,3,4,5\} \rightarrow\{3,4,5,9\}$ and $\mathrm{g}:\{3,4,5,9\}$ $\rightarrow\{7,11,15\}$ be functions defined as $\mathrm{f}(2)=3 \mathrm{f}(3)=4, \mathrm{f}(4)=\mathrm{f}(5)=5, \mathrm{~g}(3)=\mathrm{g}(4)=7$, and $g(5)=g(9)=11$. Then gof $(5)$ is equal to

**(A)** 5

**(B)** 7

**(C)** 11

**(D)** 1

**Q.3**

If $f(x)=\sqrt{3|x|-x-2}$ and $g(x)=\sin x$, then domain of definition of $\mathrm{fog}(\mathrm{x})$ is

**(A)** $\left\{2 \mathrm{n} \pi+\frac{\pi}{2}\right\}, \mathrm{n} \in \mathrm{I}$

**(B)** $\bigcup_{n \in \mathrm{I}}\left(2 \mathrm{n} \pi+\frac{7 \pi}{6}, 2 \mathrm{n} \pi+\frac{11 \pi}{6}\right)$

**(C)** $\left\{2 \mathrm{n} \pi+\frac{7 \pi}{6}\right\}, \mathrm{n} \in \mathrm{I}$

**(D)** $\left\{(4 \mathrm{~m}+1) \frac{\pi}{2}: \mathrm{m} \in \mathrm{I}\right\} \bigcup_{\mathrm{n} \in \mathrm{I}}\left[2 \mathrm{n} \pi+\frac{7 \pi}{6}, 2 \mathrm{n} \pi+\frac{11 \pi}{6}\right]$

**Q.4**

Let $f_1(x)=\begin{cases} x, 0 \leq x \leq 1 \\ 1, x>1 \\ 0, \text{ otherwise} \end{cases}$

$f_2(x)=f_1(-x)$ for all $x$

$f_3(x)=-f_2(x)$ for all $x$

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$f_4(x) = f_3(-x)$ for all $x$

Which of the following is necessarily true?

**(A)** $f_4(x) = f_1(x)$ for all $x$

**(B)** $f_1(x) = -f_3(-x)$ for all $x$

**(C)** $f_2(-x) = f_4(x)$ for all $x$

**(D)** $f_1(x) + f_3(x) = 0$ for all $x$

**Q.5**

If $f : B \to A$ is defined by $f(x) = \frac{3x+4}{5x-7}$ and $g : A \to B$ is defined by $g(x) = \frac{7x+4}{5x-3}$, where $A = R - \left\{\frac{3}{5}\right\}$ and $\mathrm{B} = \mathrm{R} - \left\{\frac{7}{5}\right\}$ and $\mathrm{I_A}$ is an identity function on $\mathrm{A}$ and $\mathrm{I_B}$ is identity function on $\mathrm{B}$, then

**(A)** $\mathrm{fog} = \mathrm{I_A}$ and $\mathrm{gof} = \mathrm{I_A}$

**(B)** $\mathrm{fog} = \mathrm{I_A}$ and $\mathrm{gof} = \mathrm{I_B}$

**(C)** $\mathrm{fog} = I_B$ and $\mathrm{gof} = I_B$

**(D)** $\mathrm{fog} = I_B$ and $\mathrm{gof} = I_A$

**Q.6**

$\mathrm{f(x)} = |\mathrm{x} - 1|, \mathrm{f} : \mathrm{R}^+ \to \mathrm{R}$ and $\mathrm{g(x)} = \mathrm{e^x}, \mathrm{g} : [-1, \infty) \to \mathrm{R}$. If the function $f(\mathrm{x})$ is defined, then its domain and range respectively are

**(A)** $(0, \infty)$ and $[0, \infty)$

**(B)** $[-1, \infty)$ and $[0, \infty)$

**(C)** $[-1, \infty)$ and $\left[1 - \frac{1}{\mathrm{e}}, \infty\right)$

**(D)** $[-1, \infty)$ and $\left[\frac{1}{\mathrm{e}} - 1, \infty\right)$

**Q.7**

If $\mathrm{f(x)} = \frac{\mathrm{x}}{\mathrm{x}-1}$, then $\frac{(\text{ fofo.....of })(\mathrm{x})}{19 \text{ times}}$ is equal to :

**(A)** $\frac{x}{x-1}$

**(B)** $\left(\frac{x}{x-1}\right)^{19}$

**(C)** $\frac{19x}{x-1}$

**(D)** $\mathrm{x}$

**Q.8**

Inverse of the function $\mathrm{f} : \mathrm{R} \to (-\infty, 1)$ given by $\mathrm{f(x)} = 1 - 2^{-\mathrm{x}}$ is

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**(A)** $-\log_2(1-x)$

**(B)** $-\log_2(x)$

**(C)** 0

**(D)** 1

**Q.9**

The inverse of $f(x) = \frac{2}{3}\frac{10^x - 10^{-x}}{10^x + 10^{-x}}$ is

**(A)** $\frac{1}{3}\log_{10}\frac{1+x}{1-x}$

**(B)** $\frac{1}{2}\log_{10}\frac{2+3x}{2-3x}$

**(C)** $\frac{1}{3}\log_{10}\frac{2+3x}{2-3x}$

**(D)** $\frac{1}{6}\log_{10}\frac{2-3x}{2+3x}$

**Q.10**

If $f(x) = x - x^2 + x^3 - x^4 + \ldots$ to $\infty$ for $|x| < 1$
then $f^{-1}(x) =$

**(A)** $\frac{x}{1+x}$

**(B)** $\frac{x}{1-x}$

**(C)** $\frac{1-x}{x}$

**(D)** $\frac{1}{x}$

**Q.11**

Let $z$ be the set of integers and $o$ be a binary operation on $z$ defined as $a \circ b = a + b - ab$ for all $a, b$ z. The inverse of an element $a(\neq 1) \in z$ is

**(A)** $\frac{a}{a-1}$

**(B)** $\frac{a}{1-a}$

**(C)** $\frac{a-1}{a}$

**(D)** None of these

**Q.12**

If $f: R \to R, fx = \frac{\sin x\pi}{x^2 + 2x + 3} + 2x - 1 + \sqrt{xx - 1 + \frac{1}{4}}$ ( where $x$ denotes greatest integral value less than or equal to $x$) denotes a function, then number of real solutions of equation $fx = f^{-1}x$ is

**(A)** 0

**(B)** 1

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**(C)** $2$

**(D)** $3$

**Q.13**

If $\mathrm{f}(\mathrm{x}+1)=\mathrm{x}^2-3\mathrm{x}+2$, then $\mathrm{f}(\mathrm{x})$ is equal to:

**(A)** $x^2-5x-6$

**(B)** $x^2+5x-6$

**(C)** $x^2+5x+6$

**(D)** $x^2-5x+6$

**Q.14**

A function $f$ is defined by $f(x)=x+\frac{1}{x}$. Consider the following.
(1) $(f(x))^2=f\left(x^2\right)+2$
(2) $(f(x))^3=f\left(x^3\right)+3f(x)$
Which of the above is/are correct?

**(A)** 1 only

**(B)** 2 only

**(C)** Both 1 and 2

**(D)** Neither 1 nor 2

**Q.15**

Let $f(x)=x^2+3x-3, x>0$. If n points $x_1, x_2, x_3, \ldots, x_n$
are so chosen on the $x$ -axis such that
(i) $\frac{1}{n} \sum_{i=1}^n f^{-1}\left(x_i\right)=f\left(\frac{1}{n} \sum_{i=1}^n x_i\right)$
(ii) $\sum_{i=1}^n f^{-1}\left(x_i\right)=\sum_{i=1}^n x_i$, where $f^{-1}$ denotes the inverse of $\mathrm{f}$. The value of $\frac{\mathrm{x}_1+\mathrm{x}_2+\ldots+\mathrm{x}_\mathrm{n}}{\mathrm{n}}=$

**(A)** 1

**(B)** 2

**(C)** 3

**(D)** 4

**Q.16**

The number of linear functions $\mathrm{f}$ satisfying $f(x+f(x))=x+f(x) \forall x \in R$ is

**(A)** 0

**(B)** 1

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**(C)** 2

**(D)** 3

**Q.17**

Iff : $\mathrm{R} \rightarrow$ Rand $g : R \rightarrow$ Rare given byf $(x) = |x|$ and $g(x) = [x]$ for each $x \in R$, then $[x \in R : g(f(x)) \leq \mathrm{f(g(x))}\} =$

**(A)** $\mathrm{Z} \cup (-\infty, 0)$

**(B)** $(-\infty, 0)$

**(C)** $\mathrm{Z}$

**(D)** $\mathrm{R}$

**Q.18**

If [.] denotes the greatest integer fimction, then the vahe of $\sum_{\mathrm{r}=1}^{100} \left[\frac{1}{2} + \frac{\mathrm{r}}{100}\right]$ is

**(A)** 49

**(B)** 50

**(C)** 51

**(D)** 52

**Q.19**

If $f(x) = \begin{cases} x, x \text{ is rational} \\ 1 - x, x \text{ is irrational} \end{cases}$ then $f(f(x))$ is

**(A)** $x \forall x \in R$

**(B)** $\begin{cases} x, x \text{ is irrational} \\ 1 - x, x \text{ is rational} \end{cases}$

**(C)** $\begin{cases} x, x \text{ is rational} \\ 1 - x, x \text{ is irrational} \end{cases}$

**(D)** None of these

**Q.20**

If $f\left(x\right) = |x - 2|$ and $g\left(x\right) = f\left(f\left(x\right)\right)$ then for $x > 10$, $\mathrm{g}'\left(\mathrm{x}\right)$ equal

**(A)** $-1$

**(B)** $0$

**(C)** $1$

**(D)** $2x - 4$

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**Q.1 Answer:**

$-1$

**Solution:**

**Given that,** $\mathrm{f}(\mathrm{x})=[\mathrm{x}]$ **and** $\mathrm{g}(\mathrm{x})=|\mathrm{x}|$
**Now,** $f\left(g\left(\frac{8}{5}\right)\right)=g\left(\frac{8}{5}\right)=\left[\frac{8}{5}\right]=1$
**and** $g\left(f\left(-\frac{8}{5}\right)\right)=g\left(\left[-\frac{8}{5}\right]\right)=g(-2)=|-2|=2$
$\therefore \mathrm{f}\left(\mathrm{g}\left(\frac{8}{5}\right)\right)-\mathrm{g}\left(\mathrm{f}\left(-\frac{8}{5}\right)\right)=1-2=-1$

**Q.2 Answer:**

11

**Solution:**

**We have gof** $(2)=g(f(2))=g(3)=7, \mathrm{gof}(3)$
$=g(f(3))=g(4)=7, \mathrm{gof}(4)=g(f(4))=g(5)=11$
**and** $\mathrm{gof}(5)=g(5)=11$

**Q.3 Answer:**

$\left\{(4\,\mathrm{m}+1)\frac{\pi}{2}:\mathrm{m}\in\mathrm{I}\right\}\bigcup_{\mathrm{n}\in\mathrm{I}}\left[2\mathrm{n}\pi+\frac{7\pi}{6}, 2\mathrm{n}\pi+\frac{11\pi}{6}\right]$

**Solution:**

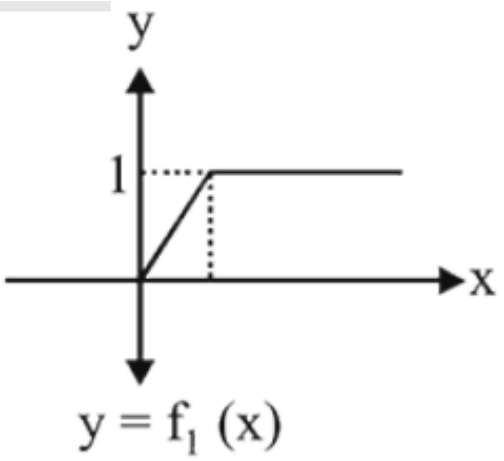
**For (fog) (x) to exists range of** $g \subseteq$ **domain of** $\mathrm{f}$.
$\therefore$ **Domain of** $\mathrm{f} \Rightarrow 3|\mathrm{x}|-\mathrm{x}-2 \geq 0$
$\Rightarrow 3|x|-x \geq 2$
**When** $x \geq 0 \Rightarrow x \geq 1$ **When** $x<0 \Rightarrow x<-\frac{1}{2}$
$\therefore \sin \mathrm{x} \geq 1$ **and** $\sin \mathrm{x}<-\frac{1}{2}$ **for** $\mathrm{f}\{\mathrm{g}(\mathrm{x})\}$ **to exists.**
**ie.,** $\sin x=1$ **and** $-1 \leq \sin x<-\frac{1}{2}$
$\therefore \mathrm{x}=(4\,\mathrm{m}+1)\frac{\pi}{2}$ **and**
$2\mathrm{n}\pi+\frac{7\pi}{6} \leq \mathrm{x} \leq 2\mathrm{n}\pi+\frac{11\pi}{6}$
**ie.,** $\left\{(4\,\mathrm{m}+1)\frac{\pi}{2}:\mathrm{m}\in\mathrm{I}\right\}\bigcup_{n\in\mathrm{I}}\left\{2\mathrm{n}\pi+\frac{7\pi}{6} \leq \mathrm{x} \leq 2\mathrm{n}\pi+\frac{11\pi}{6}\right\}$

**Q.4 Answer:**

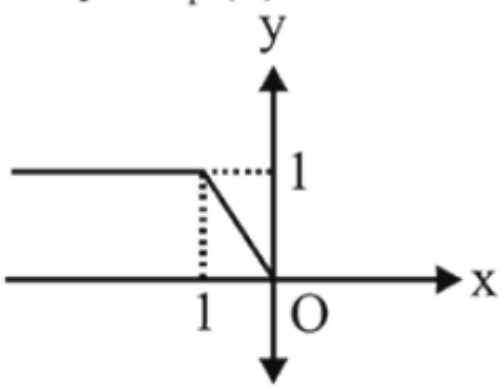
$f_1(x)=-f_3(-x)$ for all $x$

**Solution:**

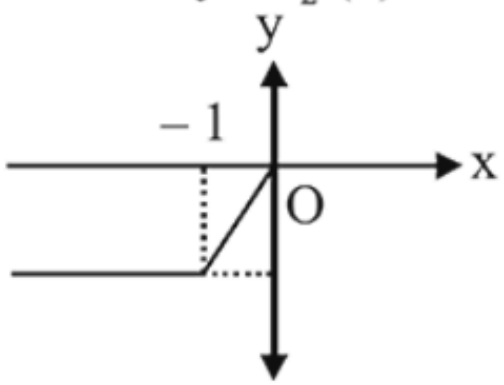
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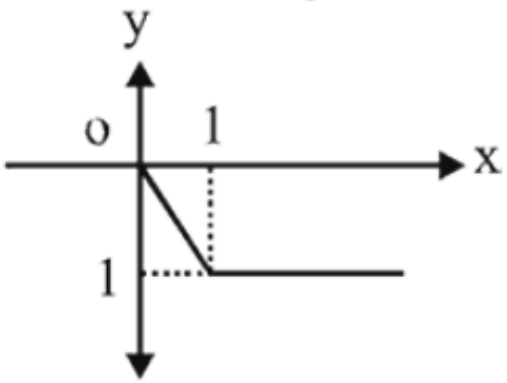

$y = f_1(x)$




$y = f_2(x)$




$y = f_3(x)$




$y = f_4(x)$

**Q.5 Answer:**

fog $= I_A$ and gof $= I_B$

**Solution:**

**We have,** $\text{gof}(x) = g\left(\frac{3x+4}{5x-7}\right) = \frac{7\left(\frac{(3x+4)}{(5x-7)}\right)+4}{5\left(\frac{(3x+4)}{(5x-7)}\right)-3}$

$= \frac{21x+28+20x-28}{15x+20-15x+21} = \frac{41x}{41} = x$

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**Similarly,** $\text{fog}(x) = f\left(\frac{7x+4}{5x-3}\right)$

$= \frac{3\left(\frac{(7x+4)}{(5x-3)}\right)+4}{5\left(\frac{(7x+4)}{(5x-3)}\right)-7}$

$= \frac{21x+12+20x-12}{35x+20-35x+21} = \frac{41x}{41} = x$

**Thus, gof** $(x) = x, \forall x \in B$ **and** $f \circ g(x) = x, \forall x \in A,$

**which implies that gof** $= \mathrm{I_B}$ **and fog** $= \mathrm{I_A}$

**Q.6 Answer:**

$[-1, \infty)$ and $[0, \infty)$

**Solution:**

$$\mathrm{f(x) = |x-1| = \begin{cases} 1-x, & 0 < x < 1 \\ x-1, & x \geq 1 \end{cases}}$$

$$\textbf{(fog)}\,(\mathrm{x}) = \begin{cases} 1-\mathrm{g(x)}, & 0 < \mathrm{g(x)} < 1 \text{ i.e. } -1 \leq \mathrm{x} < 0 \\ \mathrm{g(x)}-1, & \mathrm{g(x)} \geq 1 \text{ i.e. } 0 \leq \mathrm{x} \end{cases}$$

$$\mathrm{g(x) = e^x, x \geq -1}$$

$$\mathrm{e^x - 1, \quad x \geq 0}$$

$\therefore$ **domain** $= [-1, \infty)$

**fog is decreasing in** $[-1, 0)$ **and increasing in** $[0, \infty)$ $\mathrm{fog(-1) = 1 - \frac{1}{e}}$ **and** $\mathrm{fog(0) = 0}$

**As** $\mathrm{x \to \infty, fog(x) \to \infty}$

$\therefore$ **range** $= [0, \infty)$

$\therefore \quad \mathrm{x = \frac{1}{2} \log_e \left(\frac{y}{2-y}\right)}$

**Q.7 Answer:**

$\frac{x}{x-1}$

**Solution:**

**Given** $f(x) = \frac{x}{x-1}$

$\therefore (\textbf{ fo } f)(x) = f\{f(x)\} = f\left(\frac{x}{x-1}\right)$

$= \frac{\frac{x}{x-1}}{\frac{x}{x-1}-1} = \frac{\frac{x}{x-1}}{\frac{x-x+1}{x-1}} = \frac{\frac{x}{x-1}}{\frac{1}{x-1}} = x.$

$\Rightarrow (\textbf{ fo fo } f)(x) = f(\textbf{ fo } f)(x) = f(x) = \frac{x}{x-1}$

$\Rightarrow \underbrace{(\text{ fo fof } \ldots..0\text{f})}_{19 \text{ times}}(\mathrm{x}) = \mathrm{f}(\textbf{ fo } \mathrm{f(x)} = \mathrm{f(x)} = \frac{\mathrm{x}}{\mathrm{x-1}}$

**Q.8 Answer:**

$-\log_2(1-\mathrm{x})$

**Solution:**

**Let** $\mathrm{y = 1 - 2^{-x}}$ **or** $2^{-x} = 1 - y$ **or** $-x = \log_2(1-y)$ **or** $f^{-1}(x) = g(x)$

$= -\log_2(1-x)$

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**Q.9 Answer:**

$\frac{1}{2}\log_{10}\frac{2+3x}{2-3x}$

**Solution:**

**If** $y=\frac{2}{3}\frac{10^x-10^{-x}}{10^x+10^{-x}}$, $10^{2x}=\frac{3y+2}{2-3y}$

**or** $x=\frac{1}{2}\log_{10}\frac{2+3y}{2-3y}$

$\therefore f^{-1}(x)=\frac{1}{2}\log_{10}\frac{2+3x}{2-3x}$

**Q.10 Answer:**

$\frac{x}{1-x}$

**Solution:**

**Given** $f(x)=x-x^2+x^3-x^4+\ldots$ **to** $\infty$

$\Rightarrow y=\frac{x}{1+x}$ **( Infinite G.P. )**

$\Rightarrow y+xy=x\Rightarrow y=x(1-y)$

$\Rightarrow x=\frac{y}{1-y}$

$\Rightarrow f^{-1}(y)=\frac{y}{1-y}\Rightarrow f^{1}(x)=\frac{x}{1-x}$

**Q.11 Answer:**

$\frac{a}{a-1}$

**Solution:**

**Let** $e$ **be the identity element for the binary operation** $o$ **defined on** $z$ **given by** $a\circ b$

$=a+b-ab$

**Then** $aoe=a=eoa$ **for all** $a\in z$

$\Rightarrow a+e-ae=a$ **for all** $a\in z\Rightarrow e(1-a)=0$ **for all** $a\in z\Rightarrow e=0$

**So, 0 is the identity element for the binary operation** $o$ on $z$**.**

**Let** $x$ **be the inverse of** $a\in z$

**Then,** $a\circ x=xoa=0$

$\Rightarrow a+x-ax=0\Rightarrow x(1-a)=-a$

$\Rightarrow x=\frac{a}{a-1}(\because a\neq 1)$

**Thus,** $\frac{a}{a-1}$ **is the inverse of** $a(\neq 1)\in z$**.**

**Q.12 Answer:**

1

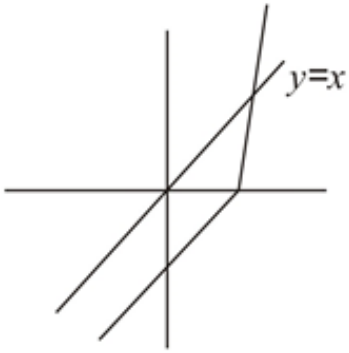
**Solution:**

$fx=2x-1+x-\frac{1}{2}$

$=\begin{matrix} x-\frac{1}{2} & x<\frac{1}{2} \\ 3x-\frac{3}{2} & x\geq\frac{1}{2}\end{matrix}$

**As** $fx$ **is increasing, solution of equation** $fx=f^{-1}x$ **is same as** $fx=x$**.**

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**∴ No. of solutions of $fx = f^{-1}x$ is one.**

**Q.13 Answer:**

$x^2 - 5x + 6$

**Solution:**

**Given function is :**

$f(x+1) = x^2 - 3x + 2$

**This function is valid for all real values of x. So, putting $x - 1$ in place of x, we get** $f(x) = f(x - 1 + 1) \Rightarrow f(x) = (x-1)^2 - 3(x-1) + 2$

$\Rightarrow f(x) = x^2 - 2x + 1 - 3x + 3 + 2$

$\Rightarrow f(x) = x^2 - 5x + 6$

**Q.14 Answer:**

Both 1 and 2

**Solution:**

$f\left(x^2\right) + 2 = x^2 + \frac{1}{x^2} + 2$

$= \left(x + \frac{1}{x}\right)^2 = \{f(x)\}^2$

**and** $f\left(x^3\right) + 3f(x)$

$= x^3 + \frac{1}{x^3} + 3\left(x + \frac{1}{x}\right)$

$= \left(x + \frac{1}{x}\right)^3 = \{f(x)\}^3$

**Thus, both 1 and 2 are correct.**

**Q.15 Answer:**

1

**Solution:**

$\frac{f^{-1}(x_1) + f^{-1}(x_2) + \ldots + f^{-1}(x_n)}{n}$

$= f\left(\frac{x_1 + x_2 + \ldots + x_n}{n}\right)$

**and** $\frac{f^{-1}(x_1) + f^{-1}(x_2) + \ldots + f^{-1}(x_n)}{n}$

$= f\left(\frac{x_1 + x_2 + \ldots + x_n}{n}\right)$

$\Rightarrow \quad f(\bar{x}) = \bar{x}$, **where** $\bar{x} = \frac{x_1 + x_2 + \ldots + x_n}{n}$

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$\Rightarrow \quad \bar{x}^2 + 3\bar{x} - 3 = \bar{x} \Rightarrow \bar{x}^2 + 2\bar{x} - 3 = 0$

$\Rightarrow \overline{\mathrm{x}} = -3, 1 \quad \Rightarrow \quad \overline{\mathrm{x}} = 1$ **as** $\overline{\mathrm{x}} > 0$

**Q.16 Answer:**

2

**Solution:**

Let $f(x) = ax + b$

$\Rightarrow \mathrm{f(ax + b + x) = x + ax + b}$

$\Rightarrow \mathrm{f((a + 1)x + b) = (a + 1)x + b}$

**Replace** $(a + 1)x + b$ **by** $y$**, we have** $\Rightarrow \mathrm{f(y) = (a + 1)\left(\frac{y-b}{a+1}\right) + b}$

$\mathrm{orf}(x) = (a + 1)\left(\frac{x-b}{a+1}\right) + b$

$\therefore$ **required no. of linear functions is** $2$**.**

**Q.17 Answer:**

R

**Solution:**

$g(f(x)) = g(|x|) = [|x|];$

$f(g(x)) = f([x]) =| [x]$

**When** $\mathrm{x \geq 0, [|x|] = [x] = |[x]|}$

$\therefore \mathrm{f(g(x)) = g(f(x))}$

**When** $x < 0, [x] \leq x < 0 \Rightarrow |[x]| \geq| x$

$\therefore \mathrm{|[x]| \geq |x| \geq [|x|]}$

$\Rightarrow \quad f(g(x)) \geq g(f(x))$

**Thus** $g(f(x)) \leq f(f(x))$ **for all** $x \in R$

**Q.18 Answer:**

51

**Solution:**

$\sum_{r=1}^{100} \left[\frac{1}{2} + \frac{r}{100}\right]$

**till** $n = 49, f(n) = 0$

**At** $n = 50, f(n) = [1] = 1$

$At_n = 100, f(n) = [3/2] = 1$

**Sum**$\rightarrow 1 + 1 + \ldots + 1$

$= 51$

**Q.19 Answer:**

$x \forall x \in R$

**Solution:**

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$$f(f(x)) = \begin{cases} f(x), & f(x) \text{ is rational} \\ 1 - f(x), & f(x) \text{ is ixrational} \end{cases}$$
$$\Rightarrow f(f(x)) = \begin{cases} x, & x \text{ is rational} \\ 1 - (1 - x) = x, & x \text{ is irrational} \end{cases}$$

**Q.20 Answer:**

1

**Solution:**

**For** $x > 10, \quad f(x) = x - 2$

**Therefore,** $g(x) = x - 2 - 2 = x - 4$

$\therefore g'(x) = 1$

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